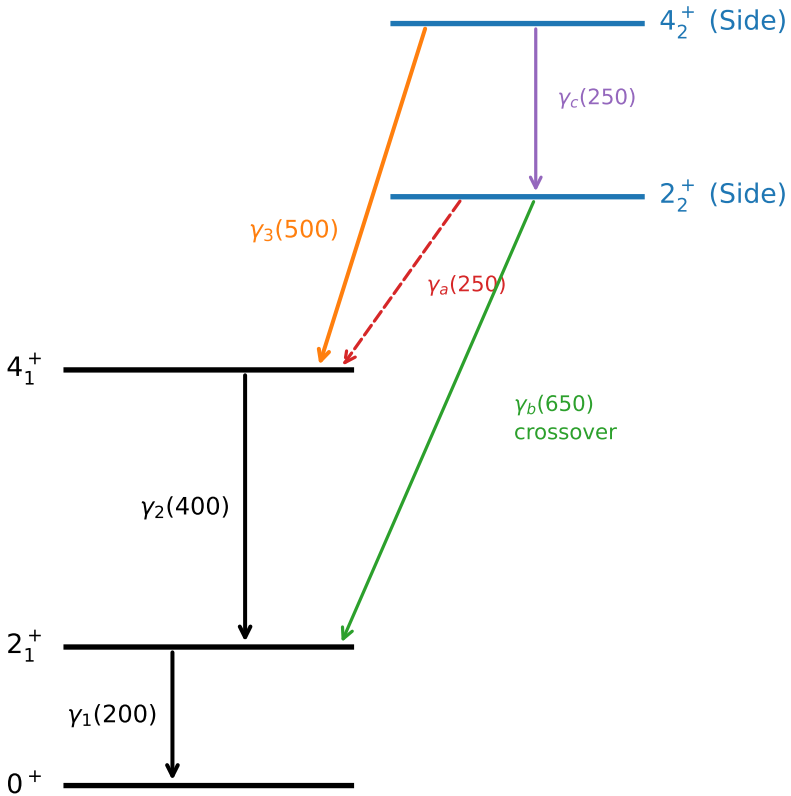
(a) Model Level Topology & Cascade Paths



(b) Gated Coincidence & Anti-coincidence Logic

Coincidence Matrix Gating Rules:

- Gate on γ_1(200 keV):

  Sees coincident γ_2(400), γ_3(500), γ_a(250), γ_b(650), γ_c(250)

- Gate on γ_2(400 keV) (Intraband):

  Sees γ_1(200), γ_3(500), γ_a(250), γ_c(250);

  Anti-coincident with crossover γ_b(650) (parallel bypass)!

- Gate on Crossover γ_b(650 keV):

  Sees γ_1(200) and γ_c(250);

  Anti-coincident with γ_2(400), γ_a(250), and γ_3(500)!

- *Multiplet Warning: γ_a & γ_c both* 250 keV *require gate differentiation.*